

# MARTHA MESLINA FLORENCIA

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Computer Science student at Universitas Padjadjaran with a focus on web application development and artificial intelligence. Experienced in developing educational chatbots, computer vision systems, and React-based user interfaces. Passionate about building interactive technologies to enhance user experience.

## EDUCATION

**PADJADJARAN UNIVERSITY**  
Computer Science

**Jatinangor, West Java**  
August 2023-Present  
(GPA : 3.82 / 4.00)

## EXPERIENCE

**LAB ASSISTANT IN DIGITAL BUSINESS UNPAD**  
**UI/UX**

**Jatinangor, West Java**  
**March 2026 - Present**

- Mentored 30+ students through the end-to-end UX process, from wireframing to high-fidelity Figma prototyping, helping them translate business ideas into user-ready digital products.
- Facilitated hands-on workshops and final project reviews with emphasis on User-Centered Design (UCD) principles and usability testing.
- Evaluated student projects against industry standards for both visual quality and business viability, providing structured feedback at each design milestone.

**CODING CAMP 2026 POWERED BY DBS FOUNDATION**  
**AI Engineer Cohort (Participant)**

**Jatinangor, West Java**  
**February 2026 - Present**

- Selected as one of 3,000 nationwide participants to intensive Artificial Intelligence and Machine Learning curriculum curated by Dicoding
- Developed and optimized machine learning models using Python, focusing on data preprocessing, feature engineering, and predictive modeling structures
- Collaborated in a cross-functional team to architect "Fingo", an AI-powered personal finance capstone project designed to predict weekly income trends and mitigate impulsive consumer buying behavior

**LAB ASSISTANT IN COMPUTER SCIENCE UNPAD**

**Data Structures, Programming Algorithm, Database System, Analysis of Algorithm**

**Jatinangor, West Java**  
**February 2025 - Present**

- Mentored 30+ students per semester in mastering core CS concepts, including Data Structures, Algorithms, and Database Systems.
- Facilitated hands-on labs for Programming Algorithms, guiding students through complex C++ implementations such as pointers, file I/O, and structured program design.
- Improved student technical proficiency by conducting code reviews and debugging sessions for various data structure operations and SQL query optimizations (Joins, Aggregation).
- Evaluated algorithm efficiency by teaching Big-O notation, time/space complexity analysis, and performance comparison of sorting/searching methods.

**PUSAT INOVASI PENGAJARAN DAN PEMBELAJARAN UNPAD**  
**Internship - Front End Developer**

**Jatinangor, West Java**  
**August 2025 - December 2025**

- Engineered an interactive AI powered learning chatbot interface using React and Three.js, enabling real-time 3D avatar rendering via WebGL
- Implemented dynamic avatar state transitions (idle, listening, thinking, speaking) synchronized with AI generated responses
- Integrated ElevenLabs Text-to-Speech API with browser fallback to deliver seamless multimodal interaction (text and voice)

## ORGANIZATIONAL EXPERIENCE

**INFORMATICS FUN DAY**  
**Project Supervisor**

**Jatinangor, West Java**  
**June 2025 - November 2025**

- Supervised multiple divisions and advised the Project Officer to keep all teams on track and meet deadlines.
- Helped make key decisions to improve teamwork and prevent project problems.

## Division Supervisor

August 2024 - November 2024

- Managed the event budget and expenses to ensure all funds were used correctly and transparently.
- Mentored division leads in event execution and creative production, fostering a cohesive workflow and ensuring all deliverables met quality standards.

## PROJECTS

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### Fingo — AI / Machine Learning | Python, FastAPI, TensorFlow, Gemini API, Streamlit

- Built a 4-week income prediction pipeline using a custom deep learning model with ResidualBlock and HuberMAE loss, deployed via FastAPI on Hugging Face Spaces.
- Integrated Gemini API to power a context-aware financial chat assistant that uses the user's own financial data as context for personalized advice.
- Iterated through LSTM, Ridge Regression, and deep learning approaches to select the best-performing architecture for gig worker income patterns.

### Papan — Web Development | Next.js, TypeScript, Prisma, PostgreSQL, NextAuth.js, Leaflet

- Built property listing, search, and user dashboard pages using Next.js App Router with full TypeScript coverage.
- Integrated Leaflet.js for interactive map-based property search and implemented a DSS recommendation UI for personalized property matching based on weighted criteria.
- Collaborated in a team with a full CI/CD pipeline using GitHub Actions and Vercel deployment.

### Urunin — Web Development | React.js, Tailwind CSS, Firebase, OCR API

- Integrated OCR API to automatically scan and parse shopping receipts, eliminating manual input for split-bill tracking.
- Built a debt tracking system with automatic per-member calculations and a personal expense dashboard with structured transaction history.
- Implemented Firebase real-time sync to keep shared expense data consistent across all users instantly.

### Zichara — Mobile Development / AR | Unity, C#, ARCore

- Designed the full AR app architecture and learning flow from character scan to interactive Hanzi and Pinyin practice.
- Developed 3D character rendering features using ARCore inside Unity for real-world overlay of Chinese writing.
- Built interactive practice pages with progress validation and UX optimized for beginner learners.

### Data Warehouse Mandiri — Data Engineering | Python, PostgreSQL, ETL, Star Schema

- Designed a Star Schema data warehouse for Bank Mandiri transaction analytics covering dimension and fact tables.
- Built incremental ETL pipelines in Python to extract, clean, and load data, with Materialized Views in PostgreSQL for efficient analytical queries.
- Connected the warehouse to Google Data Studio for an interactive business insights dashboard.

### Garbage Classification — Machine Learning / Computer Vision | Python, TensorFlow, EfficientNetV2S

- Built a CNN image classification model using EfficientNetV2S as backbone with selective layer fine-tuning for 6 waste categories.
- Applied data augmentation and class weight balancing to handle dataset imbalance and improve generalization.
- Exported the model in SavedModel, TF-Lite, and TF.js formats for flexible multi-platform deployment.

### Addicx — UI/UX Design | Figma

- Designed a community feature for peer support and a consultation flow connecting users directly with health professionals.
- Created a visual progress tracker and personalized onboarding flow based on addiction type and severity level.
- Built a complete interactive prototype from onboarding through core recovery features.

## SKILLS

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- **Programming Languages:** C++, Python, Java, C#, JavaScript, HTML, CSS.
- **Frameworks & Tools:** React.js, Next.js, Flask, Git, VS Code, Figma.
- **Soft Skills:** Communication, Team Collaboration, Time Management.